

Foundations of Adversarial machine learning

[Adversarial machine learning]

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Data collection issues Security datasets vary across formats, including binaries, network traces, and log files. Studies have reported that the process of converting these sources into features can introduce bias or inconsistencies. In addition, time-based leakage can occur when related malware samples are not properly separated across training and testing splits, which may lead to overly optimistic results.

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Adversarial examples An adversarial example refers to specially crafted input that is designed to look "normal" to humans but causes misclassification to a machine learning model. Often, a form of specially designed "noise" is used to elicit the misclassifications. Below are some current techniques for generating adversarial examples in the literature (by no means an exhaustive list).

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System design and learning Feature engineering and training can causes issues. Data snooping is a common pitfall where a model is trained using information that would not be available in a real-world scenario. Spurious correlations result when a model learns to associate artifacts with a label, rather than the underlying security-relevant pattern. For example, a malware classifier might learn to identify a specific compiler artifact instead of malicious behavior itself. Biased parameter selection is a form of data snooping where model hyperparameters are tuned using the test set.

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Byzantine attacks As machine learning is scaled, it often relies on multiple computing machines. In federated learning, for instance, edge devices collaborate with a central server, typically by sending gradients or model parameters. However, some of these devices may deviate from their expected behavior, e.g. to harm the central server's model or to bias algorithms towards certain behaviors (e.g., amplifying the recommendation of disinformation content). On the other hand, if the training is performed on a single machine, then the model is very vulnerable to a failure of the machine, or an attack on the machine; the machine is a single point of failure. In fact, the machine owner may themselves insert provably undetectable backdoors. The current leading solutions to make (distributed) learning algorithms provably resilient to a minority of malicious (a.k.a. Byzantine) participants are based on robust gradient aggregation rules. The robust aggregation rules do not always work especially when the data across participants has a non-iid distribution. Nevertheless, in the context of heterogeneous honest participants, such as users with different consumption habits for recommendation algorithms or writing styles for language models, there are provable impossibility theorems on what any robust learning algorithm can guarantee.

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Carlini & Wagner In an effort to analyze existing adversarial attacks and defenses, researchers at the University of California, Berkeley, Nicholas Carlini and David Wagner in 2016 propose a faster and more robust method to generate adversarial examples. The attack proposed by Carlini and Wagner begins with trying to solve a difficult non-linear optimization equation:

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Labeling and ground truth challenges Malware labels are often unstable because different antivirus engines may classify the same sample in conflicting ways. Ceschin et al. note that families may be renamed or reorganized over time, causing further discrepancies in ground truth and reducing the reliability of benchmarks.

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With this boundary function, the attack then follows an iterative algorithm to find adversarial examples x' for a given image x that satisfies the attack objectives.

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Iterate below
 Boundary search Gradient update Compute the gradient Find the step size
 Boundary search uses a modified binary search to find the point in which the boundary (as defined by S) intersects with the line between x and x' . The next step involves calculating the gradient for x , and update the original x using this gradient and a pre-chosen step size. HopSkipJump authors prove that this iterative algorithm will converge, leading x to a point right along the boundary that is very close in distance to the original image. However, since HopSkipJump is a proposed black box attack and the iterative algorithm above requires the calculation of a gradient in the second iterative step (which black box attacks do not have access to), the authors propose a solution to gradient calculation that requires only the model's output predictions alone. By generating many random vectors in all directions, denoted as u_b , an approximation of the gradient can be calculated using the average of these random vectors weighted by the sign of the boundary function on the image $x' + \delta u_b$, where δu_b is the size of the random vector perturbation:

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Examples Examples include attacks in spam filtering, where spam messages are obfuscated through the misspelling of "bad" words or the insertion of "good" words; attacks in computer security, such as obfuscating malware code within network packets or modifying the characteristics of a network flow to mislead intrusion detection; attacks in biometric recognition where fake biometric traits may be exploited to impersonate a legitimate user; or to compromise users' template galleries that adapt to updated traits over time. Researchers showed that by changing only one-pixel it was possible to fool deep learning algorithms. Others 3-D printed a toy turtle with a texture engineered to make Google's object detection AI classify it as a rifle regardless of the angle from which the turtle was viewed. Creating the turtle required only low-cost commercially available 3-D printing technology. A machine-tweaked image of a dog was shown to look like a cat to both computers and humans. A 2019 study reported that humans can guess how machines will classify adversarial images. Researchers discovered methods for perturbing the appearance of a stop sign such that an autonomous vehicle classified it as a merge or speed limit sign. A data poisoning filter called Nightshade was released in 2023 by researchers at the University of Chicago. It was created for use by visual artists to put on their artwork to corrupt the data set of text-to-image models, which usually scrape their data from the internet without the consent of the image creator. McAfee attacked Tesla's former Mobileye system, fooling it into driving 50 mph over the speed limit, simply by adding a two-inch strip of black tape to a speed limit sign. Adversarial patterns on glasses or clothing designed to deceive facial-recognition systems or license-plate readers, have led to a niche industry of "stealth streetwear". An adversarial attack on a neural network can allow an attacker to inject algorithms into the target system. Researchers can also create adversarial audio inputs to disguise commands to intelligent assistants in benign-seeming audio; a parallel literature explores human perception of such stimuli. Clustering algorithms are used in security applications. Malware and computer virus analysis aims to identify malware families, and to generate specific detection signatures. In the context of malware detection, researchers have proposed methods for adversarial malware generation that automatically craft binaries to evade learning-based detectors while preserving malicious functionality. Optimization-based attacks such as

GAMMA use genetic algorithms to inject benign content (for example, padding or new PE sections) into Windows executables, framing evasion as a constrained optimization problem that balances misclassification success with the size of the injected payload and showing transferability to commercial antivirus products. Complementary work uses generative adversarial networks (GANs) to learn feature-space perturbations that cause malware to be classified as benign; Mal-LSGAN, for instance, replaces the standard GAN loss with a least-squares objective and modified activation functions to improve training stability and produce adversarial malware examples that substantially reduce true positive rates across multiple detectors.